

实验：DeepWalk 算法

【环境准备】：

打开 Anaconda Prompt，创建 conda 环境 deepwalk

```
conda create --name deepwalk python=3.10
```

激活进入创建的环境 deepwalk

```
conda activate deepwalk
```

安装依赖的 Python 包

```
conda install notebook=6.5
```

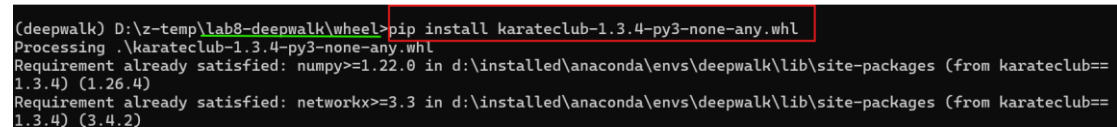
```
conda install networkx pandas numpy tqdm gensim scikit-learn matplotlib umap-learn seaborn
```

```
pip install jupyter_contrib_nbextensions
```

```
pip install ipywidgets
```

```
pip install git+https://github.com/benedekrozemberczki/karateclub.git
```

如果因网络问题无法安装 `karateclub`，则转到 `lab8-deepwalk\wheel` 路径下，执行下图命令：



```
(deepwalk) D:\z-temp\lab8-deepwalk\wheel>pip install karateclub-1.3.4-py3-none-any.whl
Processing .\karateclub-1.3.4-py3-none-any.whl
Requirement already satisfied: numpy>=1.22.0 in d:\installed\anaconda\envs\deepwalk\lib\site-packages (from karateclub==1.3.4) (1.26.4)
Requirement already satisfied: networkx>=3.3 in d:\installed\anaconda\envs\deepwalk\lib\site-packages (from karateclub==1.3.4) (3.4.2)
```

依赖的 Python 包说明：

- networkx 图网络可视化分析
- pandas 数据分析
- numpy 数据分析
- tqdm 进度条
- gensim 自然语言处理相关工具(word2vec)
- scikit-learn 机器学习(PAC 降维, TNSE 降维)

将 conda 配置到 notebook

```
conda install ipykernel
```

【任务 1】复现 DeepWalk 代码并进行可视化分析

第 1 步：获取数据

访问以下链接，按照下图步骤进行网页数据爬取。

<https://densitydesign.github.io/strumentalia-seealsology/>

The screenshot shows the web interface for strumentalia-seealsology. On the left, there is a text area for pasting Wikipedia article URLs, with three example URLs: `https://en.wikipedia.org/wiki/Deep_learning`, `https://en.wikipedia.org/wiki/Brain_science`, and `https://en.wikipedia.org/wiki/Granger_Analysis`. Below this text area is a green button labeled "START CRAWLING". To the right of the text area, there is a section for "Stop words (press enter or separate the words with a comma)". This section contains a grid of buttons for selecting different types of links to follow: "Wikipedia", "Category", "File", "Help", "Talk", "Template", "Wikisource", "Commons", "list of", "index of", "categories of", "portal", "disambiguation", and "outline of". Below this grid is a text input field for "Distance" with the value "4" and two checkboxes: "Parent links" and "Take all links". A green button labeled "DOWNLOAD" and an orange button labeled "CLEAR CACHE" are also visible.

Paste your list of wikipedia articles here or [try an example](#)

Stop words (press enter or separate the words with a comma)

Wikipedia: x Category: x File: x Help: x Talk: x Template: x

Wikisource: x Commons: x list of x index of x categories of x

portal x disambiguation x outline of x add a word and press Enter

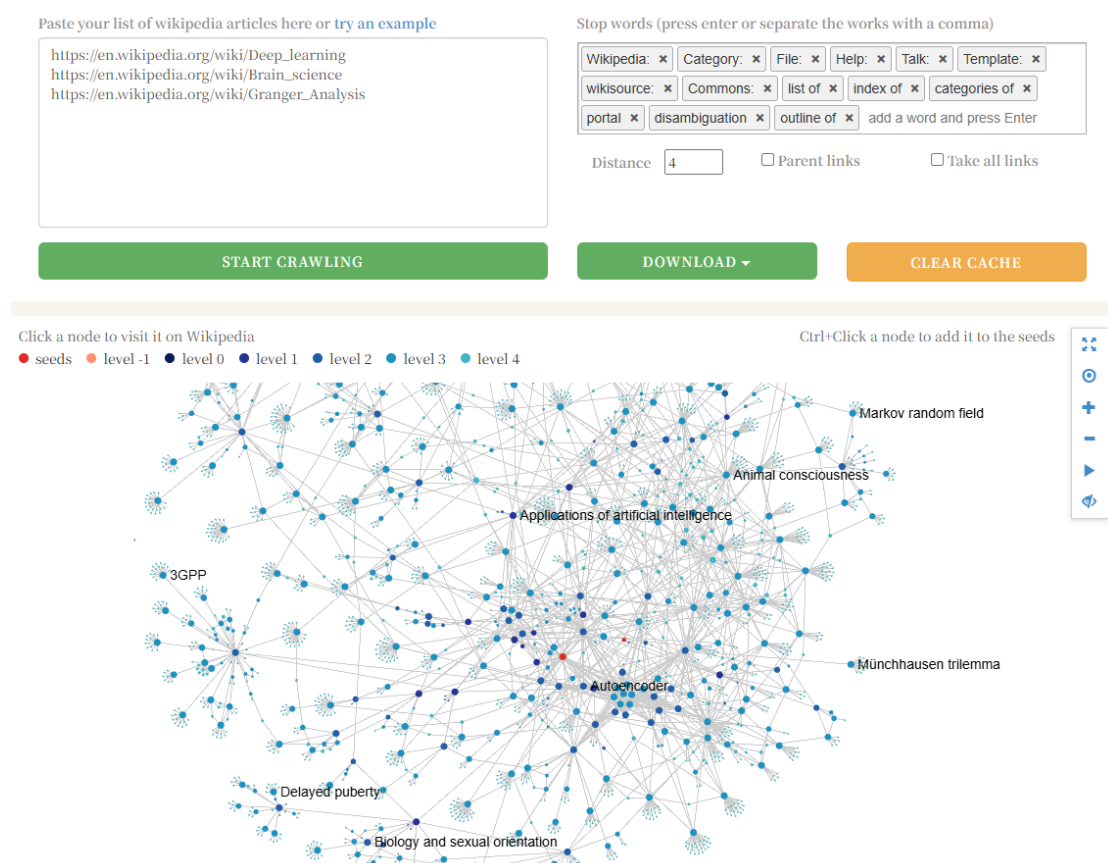
Distance 4 ☐ Parent links ☐ Take all links

START CRAWLING

DOWNLOAD

CLEAR CACHE

如下图所示，生成基于 Wikipedia 的图结构数据，并下载该数据，保存为文件 `seealsology-data.tsv`



第 2 步：运行 Notebook 脚本 `deepwalk_1.ipynb`

【粘贴截图】：请粘贴你执行 `deepwalk_1.ipynb` 脚本中的每个代码段的截图。

例如,

【代码段 1】

```
import networkx as nx

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings('ignore')

from tqdm import tqdm
import random
from gensim.models import Word2Vec
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False
```

```
plt.rcParams['font.sans-serif'] = ['SimHei']
1.4s
```

【代码段 2】

```
df = pd.read_csv('seealsoology-data.tsv', sep='\t')
df.head()
```

	source	target	depth
0	brain science	neurosemiotics	1
1	brain science	neuroscience of sex differences	1
2	brain science	gut–brain axis	1
3	brain science	connectomics	1
4	deep learning	applications of artificial intelligence	1

```
df = pd.read_csv('seealsoology-data.tsv', sep='\t')
0.0s
```

	source	target	depth
0	brain science	neurosemiotics	1
1	brain science	neuroscience of sex differences	1
2	brain science	gut–brain axis	1
3	brain science	connectomics	1
4	deep learning	applications of artificial intelligence	1

.....

【代码段 3】

```
G = nx.from_pandas_edgelist(df, 'source', 'target', edge_attr=True, create_using=nx.  
✓ 0.0s Python
```

【代码段 4】

```
return random_walk  
✓ 0.0s Python  
+ 代码 + Markdown
```

【代码段 5】

```
get_randomwalk('deep learning', 5)  
✓ 0.0s  
['deep learning',  
'cognitive architecture',  
'simulated reality',  
'blue brain project',  
'criticism of artificial neural networks']
```

【代码段 6】

```
random_walks[1]  
[10] ✓ 0.1s  
... 100%|██████████| 2671/2671 [00:00<00:00, 22143.83it/s]  
... ['brain science', 'gut-brain axis']
```

【代码段 7】

```
model.train(random_walks, total_examples=model.corpus_count, epochs=50, report_delay  
11] ✓ 4.6s Python  
.. (4807120, 4884200)
```

【代码段 8】

```
[-1.0917555 -0.12467225 -0.26825595 -0.08868978 -0.04189711 -0.07480421
-0.03645494 -0.1283342 -0.2591248 0.06905438 0.0915827 -0.03343923
0.566147 0.53479046 -0.266967 -0.05310084 -0.28018755 0.24795721
0.18008561 -0.11307164 -0.39963526 0.2646602 -0.328623 0.23016772
-0.59852386 -0.10348944 0.31360447 0.06608903 0.19734618 0.35974267
-0.00763172 -0.5418606 0.02533618 0.22141346 -0.03032703 -0.11558994
0.04405083 0.2848465 -0.18782803 -0.11355576 0.5863095 -0.42340642
-0.15318598 -0.10921836 0.23718672 -0.2847025 0.23017257 -0.09990627
-0.04540505 0.2913406 -0.1417861 0.10461096 0.02892357 0.19133927
0.65680045 -0.0988784 -0.0332135 -0.48300952 -0.09844077 -0.08625294
0.20102762 -0.3059376 0.15804268 -0.04026339 -0.6625001 -0.1976894
-0.36804572 -0.07570873 -0.5393521 -0.19247322 0.14383364 -0.10485329
-0.17403167 0.34055346 0.36532825 0.1494288 -0.2003632 -0.28210366
-0.10709383 -0.5942806 0.01884849 0.05579448 -0.27795815 0.70156205
-0.373172 0.6996889 -0.22362803 0.1280209 -0.48778522 -0.27303717
0.37468415 0.33286923 -0.18624422 0.26913336 0.80538076 -0.19068752
0.1862469 -0.3597435 -0.5726506 0.15325831 0.1497856 0.2630574
0.5055467 0.3304532 0.6227656 0.6266491 -0.13651416 -0.18255448
-0.00565149 0.48652238 0.09781732 -0.16962498 0.0182788 -0.19970983
-0.20244358 0.10101029 0.15783703 0.21239279 0.2159525 0.37384185
0.23171549 -0.23624232 0.27443755 0.6011543 0.14043246 -0.06830771
-0.14905508 -0.11984298 0.747481 0.13904326 0.3761452 0.51335466
0.38340744 -0.6793895 -0.0322228 0.31667536 -0.23636945 -0.02459503
-0.23155825 0.39947802 -0.18534407 0.6852678 0.05748295 0.28965095
-0.4806217 -0.08505888 -0.4984173 0.44965324 0.3726741 -0.2561653
...
-0.9872276 -0.16136624 -0.02102545 -0.34057462 0.93115485 0.6005495
-0.29809687 -0.54060876 -0.2726544 -0.4829954 0.05554772 -0.3341002
0.20019531 0.22348222 0.37425044 0.41817322]
(['gut-brain axis', 0.9014146327972412), ('neurosemiotics', 0.8045030236244202), ('sexually dimc
```

【代码段 9】

```
# 导入节点的嵌入向量
X = model.wv.vectors

pca = PCA(n_components=2) # 输入2维
embed_2d = pca.fit_transform(X)

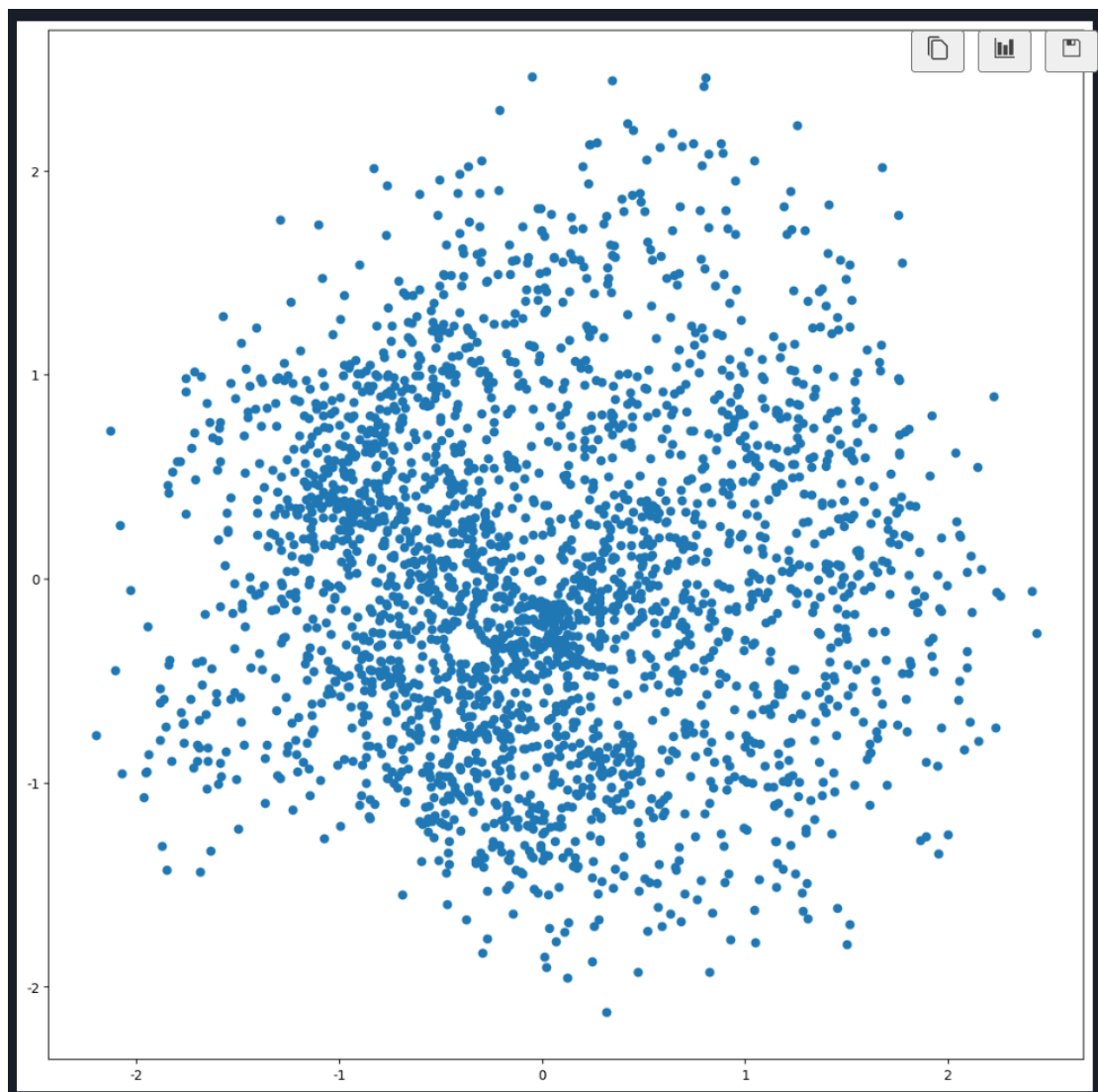
embed_2d.shape
✓ 0.0s
(2671, 2)
```

【代码段 10】

```
plt.figure(figsize=(14,14))
plt.scatter(embed_2d[:, 0], embed_2d[:, 1])
plt.show()
```

✓ 0.1s Python

findfont: Generic family 'sans-serif' not found because none of the following families were found
findfont: Generic family 'sans-serif' not found because none of the following families were found
findfont: Generic family 'sans-serif' not found because none of the following families were found
findfont: Generic family 'sans-serif' not found because none of the following families were found
findfont: Generic family 'sans-serif' not found because none of the following families were found



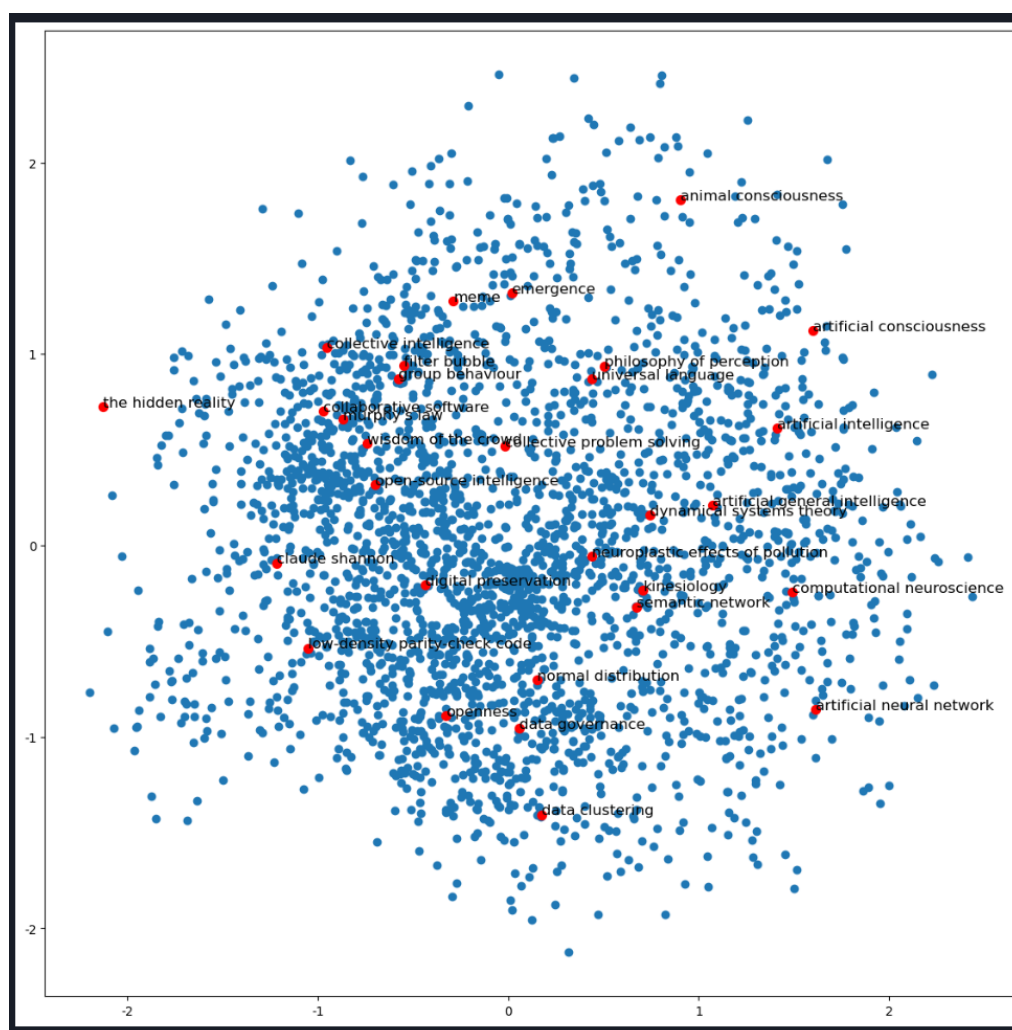
【代码段 11】

```
# 输入词条，输出词典中的索引号
term2index = model.wv.key_to_index

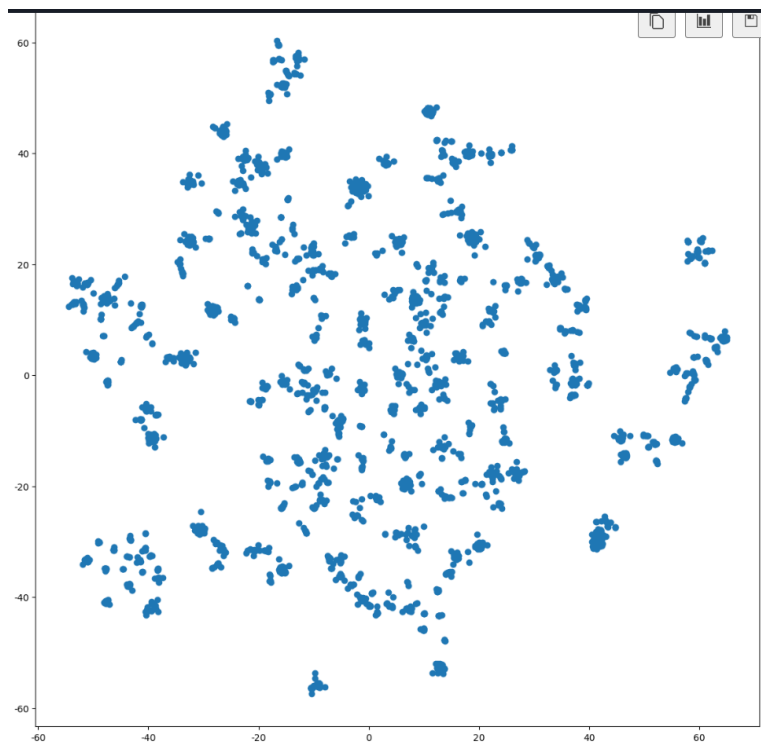
5] ✓ 0.0s
```

【代码段 12】

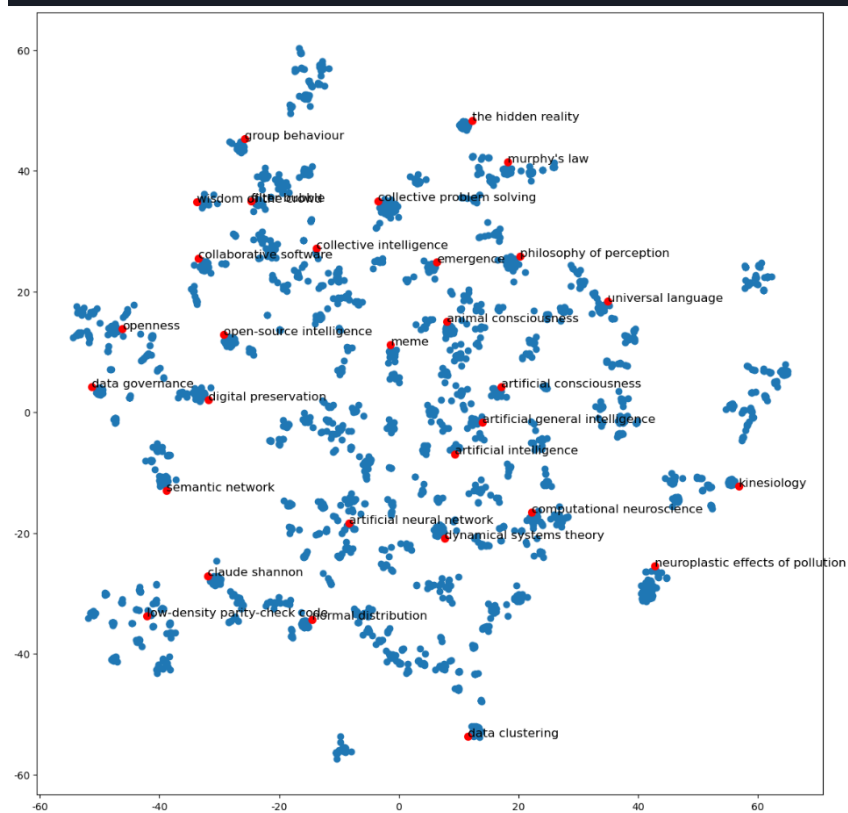
```
✓ 0.2s Python
findfont: Generic family 'sans-serif' not found because none of the following families were four
findfont: Generic family 'sans-serif' not found because none of the following families were four
findfont: Generic family 'sans-serif' not found because none of the following families were four
findfont: Generic family 'sans-serif' not found because none of the following families were four
```



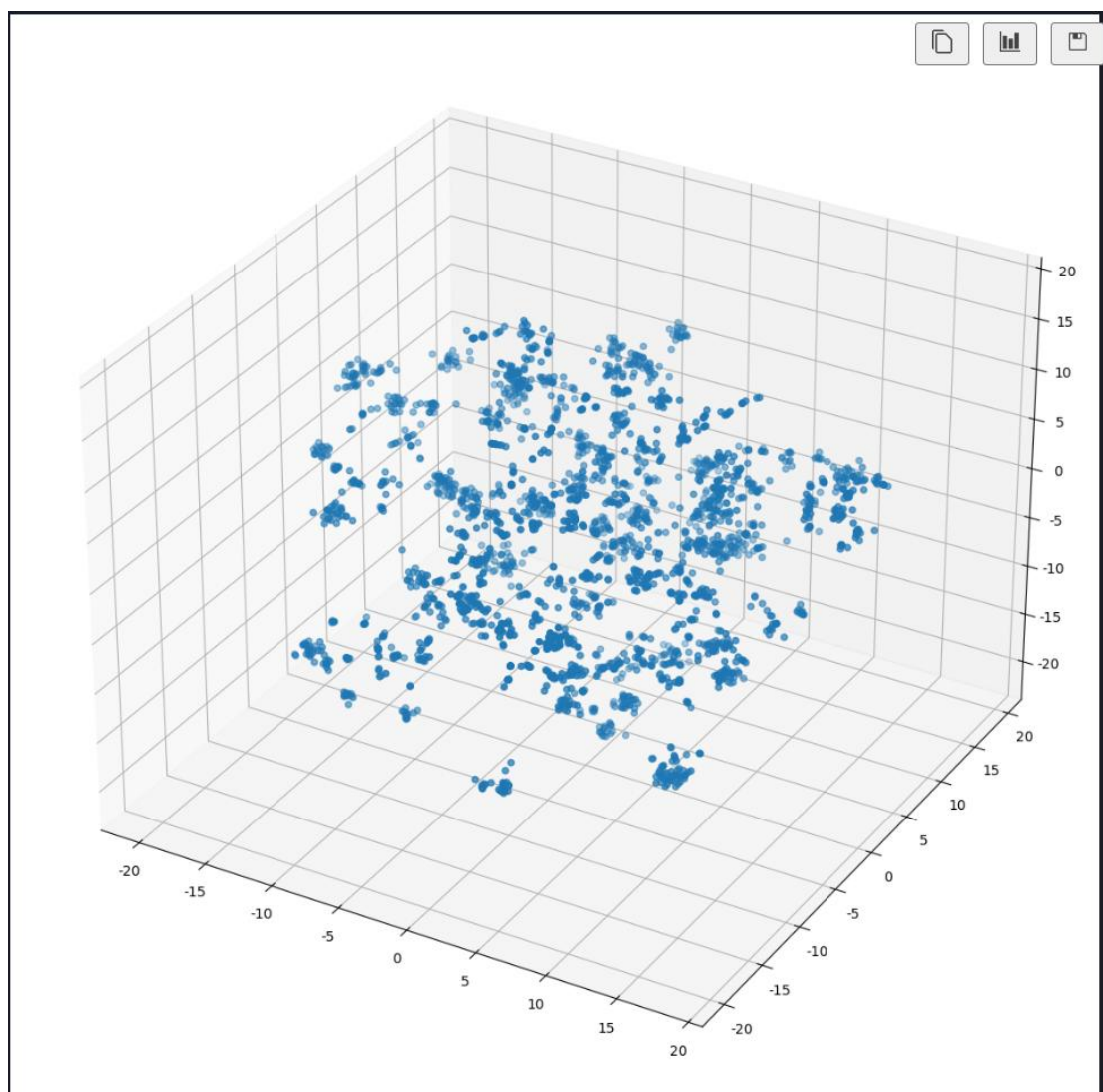
【代码段 13】



【代码段 14】



【代码段 15】



【任务 2】复现 DeepWalk 与 Node2Vec 代码并可视化

运行 Notebook 脚本 `deepwalk_2.ipynb`

【粘贴截图】：请粘贴你执行 `deepwalk_2.ipynb` 脚本中的每个代码段的截图。

例如，

【代码段 1】

```

import networkx as nx
import json

import umap
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib as mpl

import pandas as pd
import numpy as np
from tqdm import tqdm
import random
from scipy.linalg import sqrtm

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score, f1_score, confusion_matrix

from karateclub.utils.walker import RandomWalker, BiasedRandomWalker
from karateclub import DeepWalk, Node2Vec
from gensim.models.word2vec import Word2Vec

```

```

from gensim.models.word2vec import Word2Vec

```

8.6s

【代

码段 2】

```

%matplotlib inline
mpl.rcParams['figure.figsize'] = 18, 7
pd.set_option('display.float_format', lambda x: '%.5f' % x)

```

0.0s

【代码段 3】

```

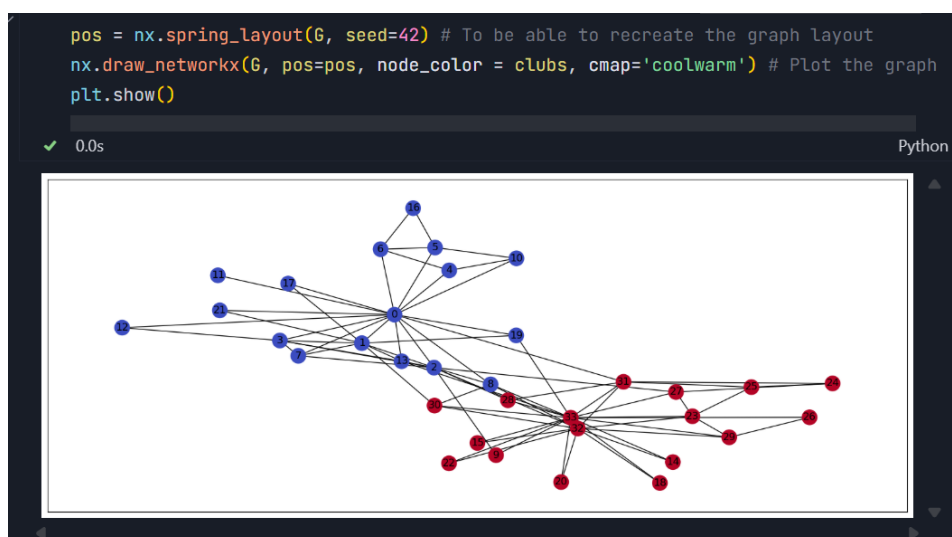
G = nx.karate_club_graph() # load data

clubs = [] # list to populate with labels
for n in G.nodes:
    c = G.nodes[n]['club'] # karate club name, can be either 'Officer' or 'Mr. Hi
    clubs.append(1 if c == 'Officer' else 0)

```

0.0s

【代码段 4】



【代码段 5】

```
model = DeepWalk(dimensions=124) # node embedding algorithm
model.fit(G) # fit it on the graph
embedding = model.get_embedding() # extract embeddings

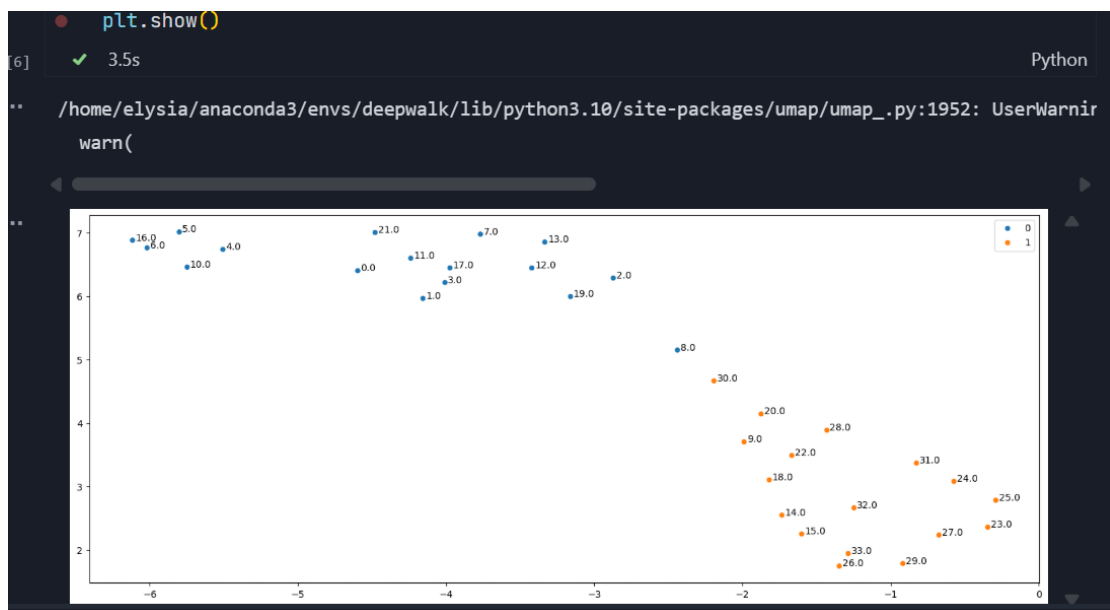
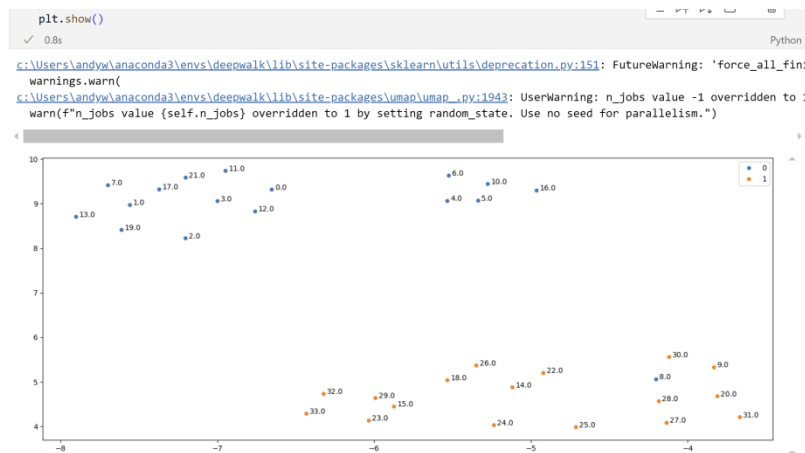
print('Number of karate club members:', len(G.nodes))
print('Embedding array shape:', embedding.shape)
```

0.0s

Number of karate club members: 34
Embedding array shape: (34, 124)

/home/elysia/anaconda3/envs/deepwalk/lib/python3.10/site-packages/karateclub/estimator.py:66
warnings.warn(

【代码段 6】



【代码段 7】

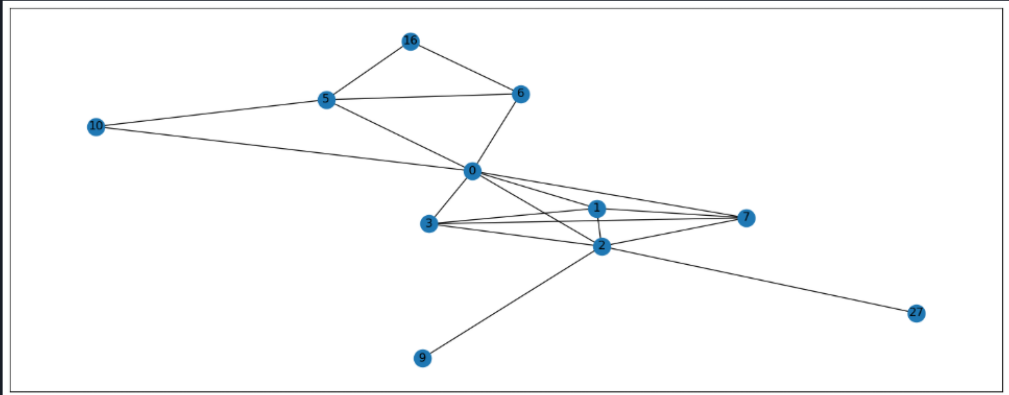
```
return walk
```

✓ 0.0s Python

【代码段 8】

```
[8] ✓ 0.0s Python
... Steps in random walk: [6, 5, 16, 5, 10, 5, 16, 5, 0, 7, 1, 3, 2, 27, 2, 7, 3, 2, 9, 2, 1]
/home/elysia/anaconda3/envs/deepwalk/lib/python3.10/site-packages/networkx/drawing/nx_pyplot.py:10:
node_collection = ax.scatter(
...

```



【代码段 9】

```
walker = RandomWalker(walk_length = 80, walk_number = 10)
walker.do_walks(6) # you can access the walks in walker.walks
```

[9] ✓ 0.0s Python

【代码段 10】

```
seed=42)
```

[10] ✓ 0.0s Python

【代码段 11】

```
print('Shape of Embedding matrix: ', embedding.shape)
```

[11] ✓ 0.0s

```
... Shape of embedding matrix: (34, 128)
```

And that's it! The embeddings are trained, so you can use them e.g. as features for your supervised learning task.

【代码段 12】

```
[12] features_path = 'twitch/PTBR/musae_PTBR_features.json' ✓ 0.0s
```

【代码段 13】

```
[13] ✓ 0.0s
```

...	from	to
0	0	92
1	0	428
2	1	689
3	1	1147
4	1	1666

【代码段 14】

```
[14] ✓ 0.0s
```

...		id	days	mature	views	partner	new_id
	new_id						
	1706	44891403	1943	False	929459	True	1706
	1273	61180621	1633	False	11194	False	1273
	313	145109685	632	False	2850	False	313
	1570	122121521	906	True	3422	False	1570
	800	189445819	267	False	71	False	800

【代码段 15】

```
[15] ✓ 0.0s
```

[+ 代码](#) [+ Markdown](#)

【代码段 16】

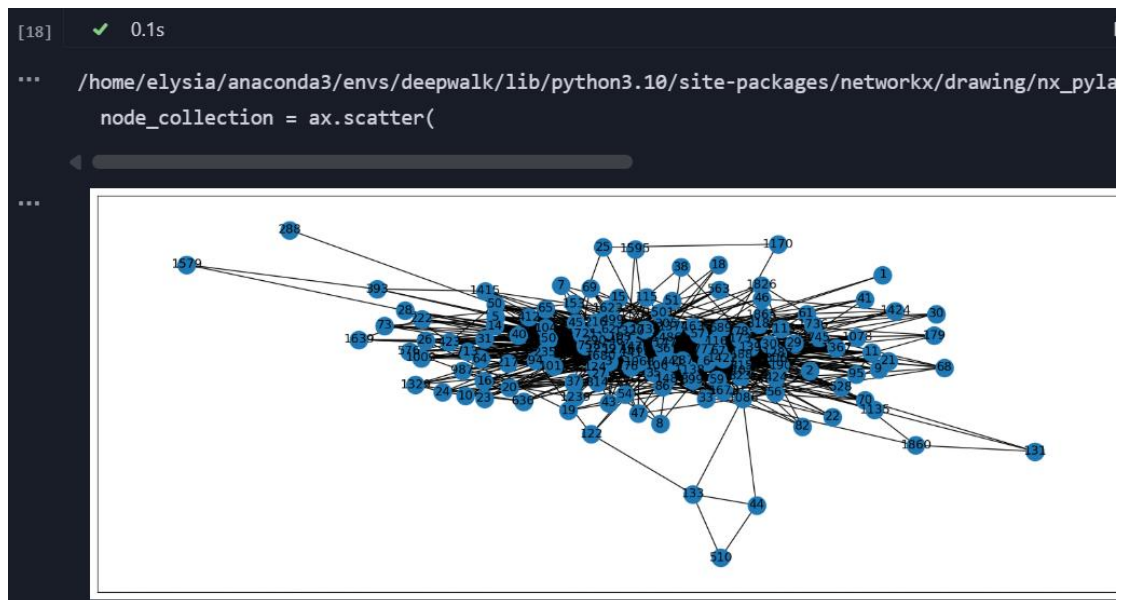
```
] ✓ 0.0s
```

100%|██████████| 1912/1912 [00:00<00:00, 75406.76it/s]

【代码段 17】

[17] ✓ 0.0s Python

【代码段 18】



【代码段 19】

```
walker = RandomWalker(walk_length = 80, walk_number = 10)
walker.do_walks(graph)
```

[19] ✓ 3.8s

【代码段 20】

[20] ✓ 4.9s

```
... Exception ignored in: 'gensim.models.word2vec_inner.our_dot_float'
Exception ignored in: 'gensim.models.word2vec_inner.our_dot_float'
Exception ignored in: 'gensim.models.word2vec_inner.our_dot_float'
Exception ignored in: 'gensim.models.word2vec_inner.our_dot_float'
Exception ignored in: 'gensim.models.word2vec_inner.our_dot_float'
```

【代码段 21】

[21] ✓ 0.0s

...

	id	days	mature	views	partner	new_id
new_id						
86	34844857	2248	False	108200	False	86
843	146094379	623	False	9530	False	843
554	68760758	1522	False	299572	True	554
1227	27535961	2398	False	1390	False	1227
268	130339417	810	True	4637	False	268

【代码段 22】

[22] ✓ 0.0s

...

【代码段 23】

[23] ✓ 0.9s

...

0.6266318537859008

[[0.95319149 0.04680851]

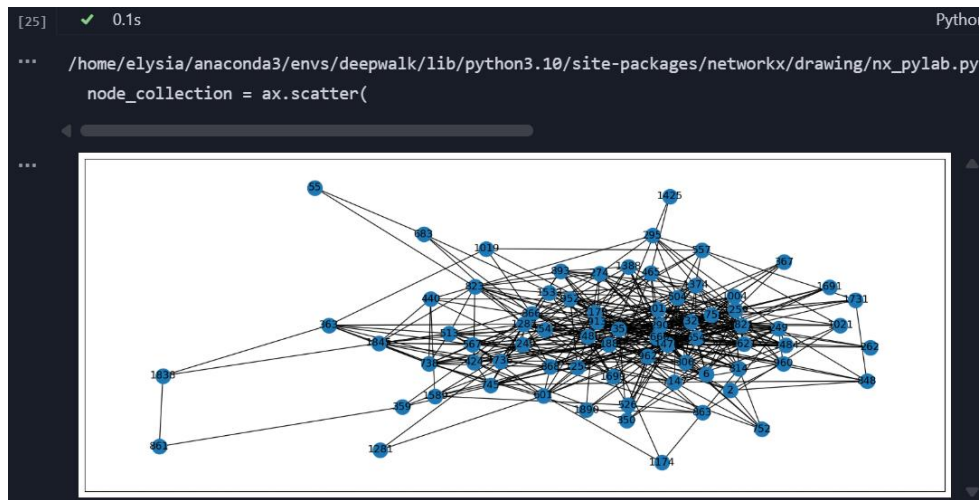
[0.89189189 0.10810811]]

【代码段 24】

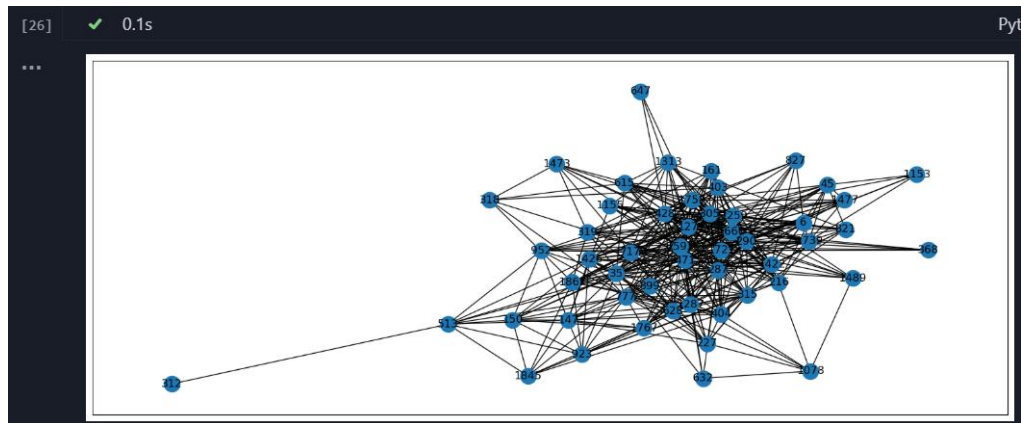
[24] ✓ 0.0s

...

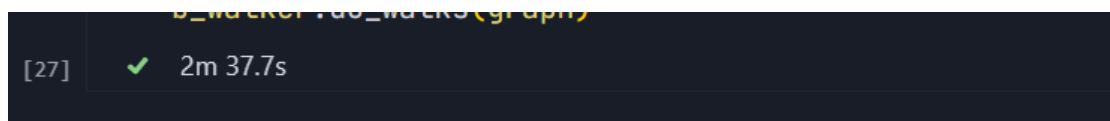
【代码段 25】



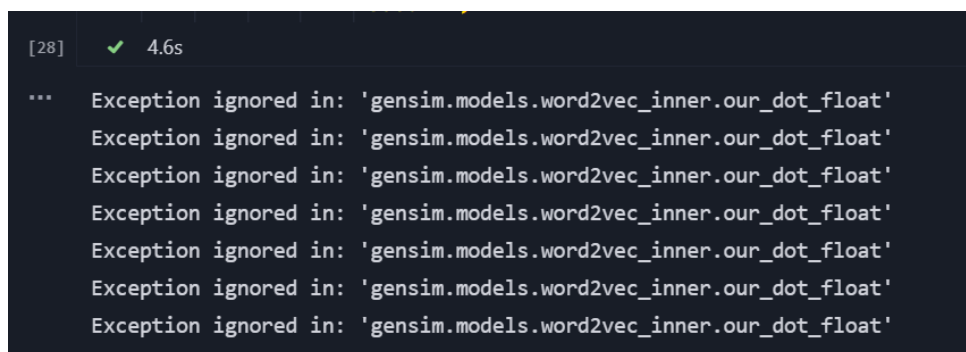
【代码段 26】



【代码段 27】



【代码段 28】



【代码段 29】

```
[29] ✓ 0.0s
```

【代码段 30】

```
[30] ✓ 0.9s  
... 0.6736292428198434  
    [[0.95967742 0.04032258]  
     [0.85185185 0.14814815]]
```

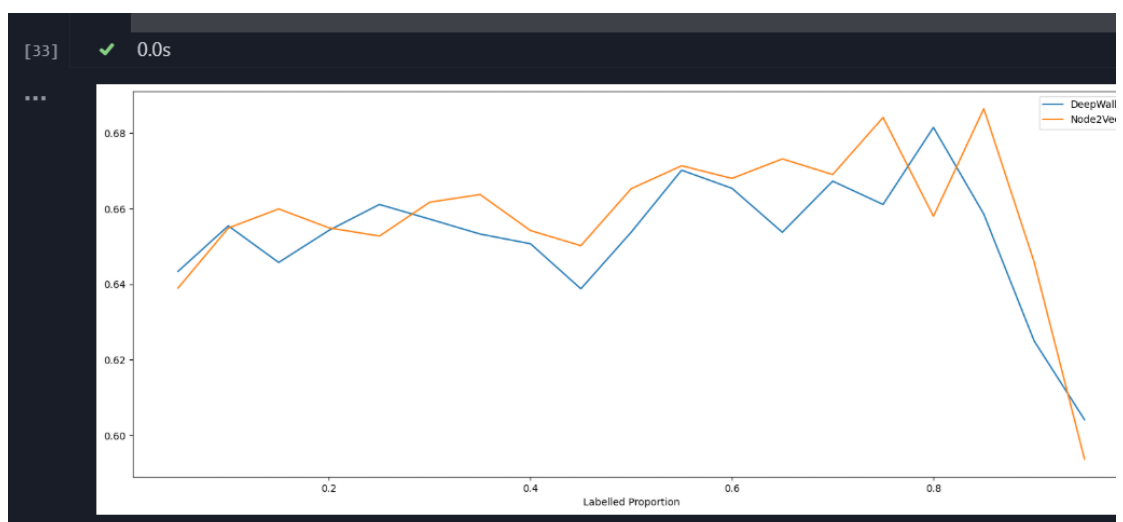
【代码段 31】

```
[31] ✓ 10.6s Python  
... 100%|██████████| 19/19 [00:10<00:00, 1.78it/s]
```

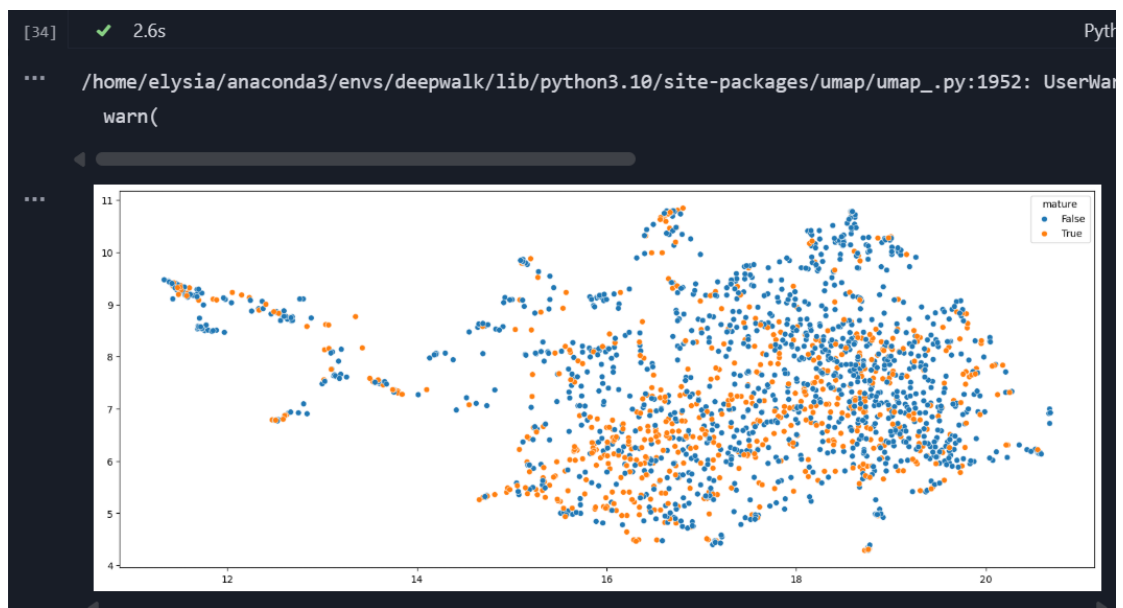
【代码段 32】

```
nv_macro_f1_scores.append(mac)  
[32] ✓ 10.5s  
... 100%|██████████| 19/19 [00:10<00:00, 1.80it/s]
```

【代码段 33】



【代码段 34】



【代码段 35】

